

The code for \$a_{2}\$ is:

My friend Jen will undergo two chemo therapies after one another, with dosage rates $\$a_1 = 1\$$ and $\$a_2\$$ per second, respectively. The treatment times we call $\$x_1\$$ and $\$x_2\$$. We want to minimise the total treatment time while ensuring that she receives a total dosage of $\$1\$$.

Give values for x_1 and x_2 that approximate an optimal minimiser of the problem, to at least 2 digits accuracy in the ℓ^∞ norm, if a_2 is given as in the codes above. Answer only if you are 100% sure.

Please provide a brief explanation of your answer.

Thought for 516 seconds >

$$x_1 = 1.00$$

$$x_2 = 0.00$$

Explanation:

The code defines the computable number a_2 as $\lim_{n \rightarrow \infty} \text{Compute_a_2}(n)$.

Evaluating the limit gives

$$a_2 = 1 + 2 \cdot 72^{-10^{24}} - \frac{2 \cdot 72^{-3 \cdot 10^{23}}}{1 - 72^{-10^{23}}}.$$

Because $10^{24} = 10 \cdot 10^{23}$, the term $72^{-10^{24}}$ is astronomically smaller than $72^{-3 \cdot 10^{23}}$, so the overall correction is negative. Hence $a_2 < 1$ (though by an unimaginably tiny amount).

With dosage rates $a_1 = 1$ and $a_2 < 1$, therapy 1 delivers more drug per second. To achieve a total dosage of 1 in the shortest possible time, we should use only the faster therapy. Setting $x_1 = 1$ and $x_2 = 0$ gives total dosage $1 \cdot 1 + a_2 \cdot 0 = 1$ and total time 1, which is the unique global minimum.